

Expt. No. : 13

Date :

SCATTER PLOT

AIM

To write a python program to draw the scatter plot using matplotlib.

SOFTWARE REQUIRED

Python 3.14

THEORY

SCATTER PLOT:

The scatter plots are mostly used for comparing variables when we need to define how much one variable is affected by another variable. The data is displayed as a collection of points. Each point has the value of one variable, which defines the position on the horizontal axes, and the value of other variable represents the position on the vertical axis.

The basic syntax for creating a scatter plot in Python using matplotlib is:

```
plt.scatter(x, y, s=None, c=None, marker=None, cmap=None)
```

Key Parameters:

- x: List or array-like, x-coordinates of points.
- y: List or array-like, y-coordinates of points.
- s: Size of the markers (optional).
- c: Color of the markers (optional).
- marker: Shape of the markers (optional).
- cmap: Colormap for mapping marker colors (optional, used with c).

PROBLEM

1. Build a Python program to create a scatter plot showing the relationship between daily exercise time and calories burned.
2. Build a Python program to create a scatter plot for the drain characteristic of a JFET with $V_{GS} = 0\text{ V}$ by plotting drain-source voltage (V_{DS}) on the X-axis and drain current (I_D) on the Y-axis.
3. For a given integer K, print the first K rows of Pascal's Triangle. Print each row with each value separated by a single space. The value at the nth row and rth column of the triangle is equal to $n!/(r!(n-r)!)$ where indexing starts from 0. These values are the binomial coefficients.

ALGORITHM

STEP 1: Start the program.

STEP 2: Open a file in import matplotlib.pyplot.

STEP 3: Get the input for x values using the input() function.

STEP 4: Get the input for y values using the input() function.

STEP 5: Write the x and y values into the file.

STEP 6: plot the value according to the given value in a plt.scatter using the x and y

values.

STEP 7: Finally show the plotted value in the graph.

STEP 8: Close the file, stop the program.

PROGRAM CODING

```

import matplotlib.pyplot as plt
n = int(input("Enter number of readings: "))
time = []
calories = []
for i in range(n):
    t = float(input("Enter exercise time: "))
    c = float(input("Enter calories burned: "))
    time.append(t)
    calories.append(c)
plt.scatter(time, calories)
plt.xlabel("Exercise Time")
plt.ylabel("Calories Burned")
plt.title("Exercise Time vs Calories")
plt.show()

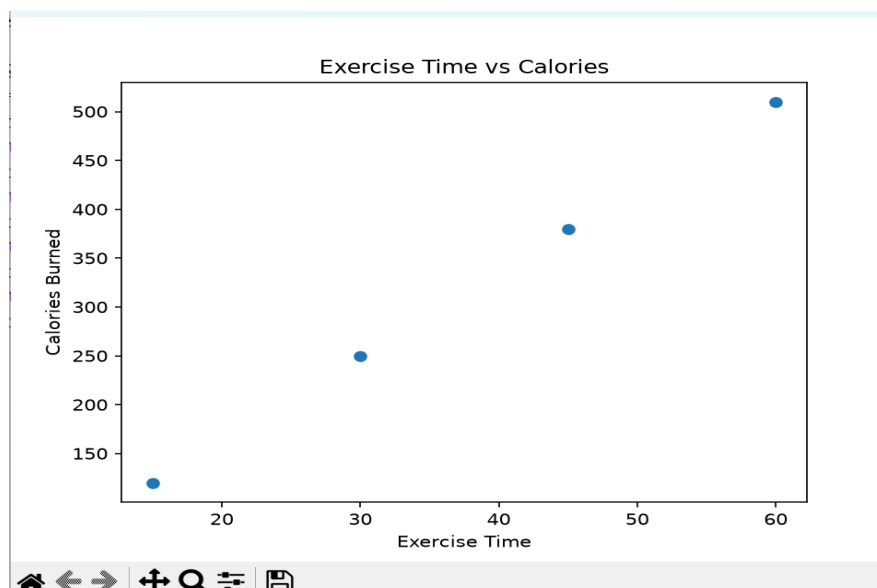
```

OUTCOMES OF PROGRAM CODING

```

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.194
4 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Experiment\Experi
ment13.py =====
Enter number of readings: 4
Enter exercise time: 15
Enter calories burned: 120
Enter exercise time: 30
Enter calories burned: 250
Enter exercise time: 45
Enter calories burned: 380
Enter exercise time: 60
Enter calories burned: 510
>>>

```



PROBLEM CODING

```

import matplotlib.pyplot as plt
n = int(input("Enter number of readings: "))
vds = []
id_value = []
for i in range(n):
    v = float(input("Enter VDS: "))
    current = float(input("Enter ID: "))
    vds.append(v)
    id_value.append(current)
plt.scatter(vds, id_value)
plt.xlabel("VDS")
plt.ylabel("ID")
plt.title("JFET Drain Characteristic (VGS = 0V)")
plt.show()

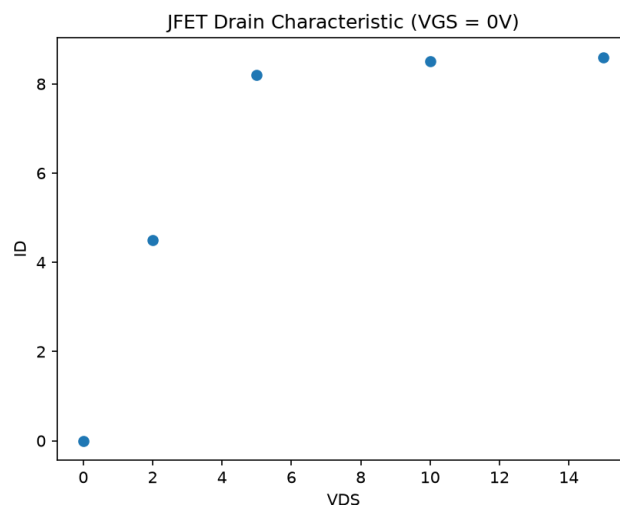
```

OUTCOMES OF PROGRAM CODING

```

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)]
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Problem\experiment13.py =====
Enter number of readings: 5
Enter VDS: 0
Enter ID: 0
Enter VDS: 2
Enter ID: 4.5
Enter VDS: 5
Enter ID: 8.2
Enter VDS: 10
Enter ID: 8.5
Enter VDS: 15
Enter ID: 8.6

```

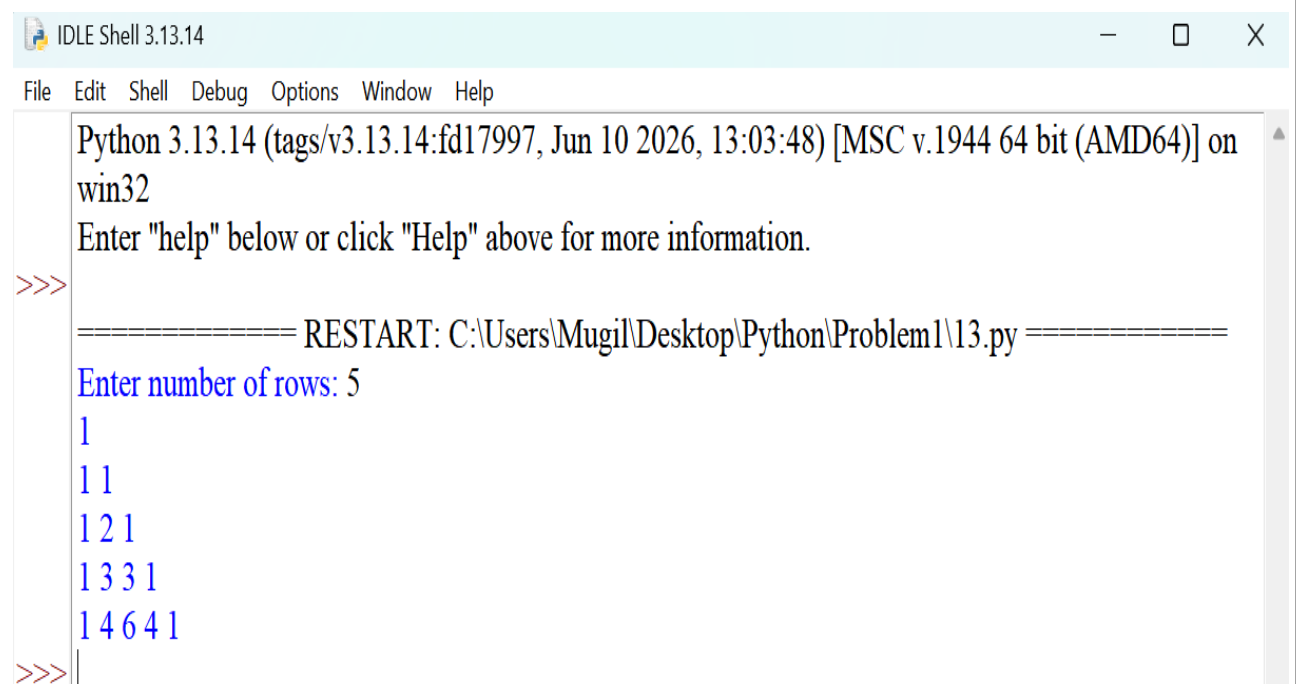


PROBLEM CODING

```

k = int(input("Enter number of rows: "))
for n in range(k):
    value = 1
    for r in range(n + 1):
        print(value, end=" ")
        value = value * (n - r) // (r + 1)
    print()

```

OUTCOMES OF PROGRAM CODING


Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32

Enter "help" below or click "Help" above for more information.

```

>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Problem1\13.py =====
Enter number of rows: 5
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
>>>

```

VIVA QUESTION**1. What is Matplotlib in Python?**

Matplotlib is a popular Python library used for creating static, interactive, and animated visualizations. It is widely used for generating plots like line charts, scatter plots, bar charts, histograms, and more.

2. How do you install Matplotlib?

You can install Matplotlib using the following pip command: `pip install matplotlib`

3. What is the difference between `plt.plot()` and `plt.scatter()`?

`plt.plot()` is used to create a line plot, where points are connected by lines. `plt.scatter()` is used to create a scatter plot, which displays individual points without connecting them.

4. What is the use of `plt.xlabel()` and `plt.ylabel()`?

`plt.xlabel()` is used to label the X-axis, and `plt.ylabel()` is used to label the Y-axis. They help in identifying what each axis represents in the plot.

5. How can you add a title to a plot in Matplotlib?

You can add a title to a plot using the `plt.title()` function. For example:
`plt.title('Sample Plot Title')`

6. What is the purpose of `plt.legend()`?

`plt.legend()` is used to display a legend on the plot, which helps in identifying different data series or categories within the plot when multiple lines or sets of points are plotted.

7. How can you change the color of the plot markers in Matplotlib?

You can change the color of markers in a scatter plot by using the `c` parameter. For example: `plt.scatter(x, y, c='red')`

8. What is a bar plot, and how is it created in Matplotlib?

A bar plot displays categorical data with rectangular bars representing the frequency or value of each category. It can be created using `plt.bar()`:
`plt.bar(categories, values)`

9. What is a histogram, and when would you use it?

A histogram is used to display the distribution of a dataset by dividing it into intervals (bins). It shows how frequently values fall into each bin. You can create a histogram with `plt.hist()`: `plt.hist(data, bins=10)`

10. How do you display grid lines on a plot in Matplotlib?

You can display grid lines on a plot using the `plt.grid()` function. For example:
Python `plt.grid(True)`

RESULT:

Thus, the Python program successfully generated a scatter plot using the given data. The relationship between the two variables was represented graphically, and the plotted data points were displayed and verified successfully.

INFERENCE:

The scatter plot provides a simple visual representation of the relationship and variation between two variables. It helps identify patterns, trends, correlations, and possible outliers in the given dataset, supporting effective data analysis and interpretation.

Assessment	Max. Marks	Marks Obtained
Conduction of Practical	15	
Report	5	
Viva	5	
Total Marks	25	
Signature of Lab In-Charge		